



THE UNIVERSITY OF
MELBOURNE

Machine Learning Applications for Health

COMP90089 (2026) - Lecture 1
Prof Daniel Capurro





About COMP90089 - Staff

Prof Daniel Capurro (Subject Coordinator)

School of Computing and Information Systems

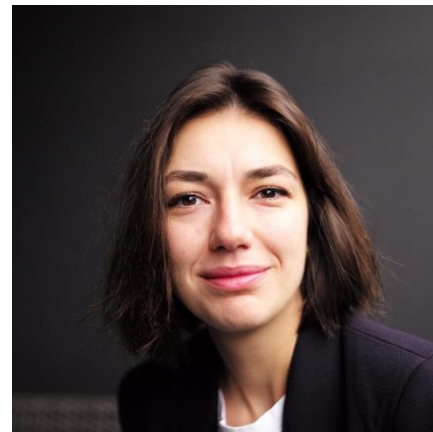
Royal Melbourne Hospital

Peter MacCallum Cancer Centre



Dr. Vlada Rozova (Lecturer)

School of Computing and Information Systems





About COMP90089 - Tutors

Roben Delos Reyes (r.delosreyes@unimelb.edu.au)

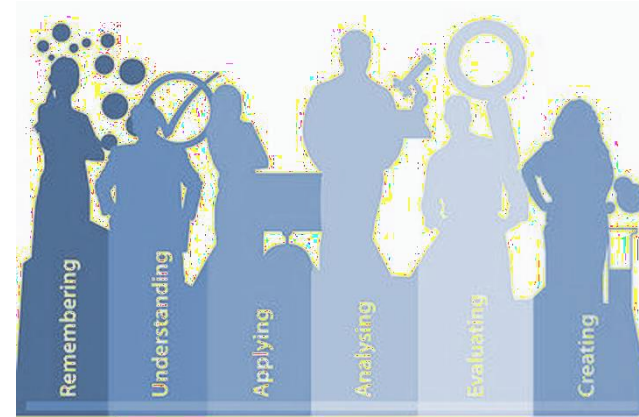
Jemima Kang (jemima.kang.1@unimelb.edu.au)

Jie Huang (jie.huang.4@unimelb.edu.au)

Learning Outcomes

On completion of this subject, students should be able to:

- Describe the application of Machine Learning concepts in the context of health problems
- Demonstrate familiarity with challenges associated with health data and data modelling
- Design, implement, and evaluate a Machine Learning system addressing a healthcare problem
- Critically assess Machine Learning applications for health





About COMP90089 - Assessment

Regular (in class) quizzes worth 20% in total

- 3-4 questions covering the week's content

Two programming assignments worth 30% in total

- Assignment 1 - 15%
- Assignment 2 - 15%
 - 50% hurdle collectively
- Assignments/exam will require some programming
 - A mixture of SQL, Python, pseudocode and “explain with your own words”

Group Project worth 10% (proposal) + 40% (final report)

- 50% hurdle
- 4-5 students in a group by week 3 - group contract to be signed

Organisations

This subject is offered **on campus only**

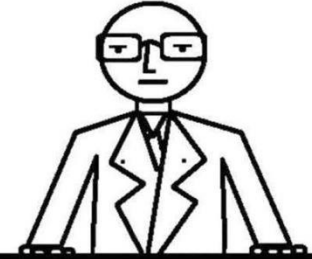
- The lectures are **on campus**
- Tutorials are **on campus**
- All lectures will be recorded. All recordings and other materials will be made available online through Canvas
- Attending the lectures and tutorials is **highly recommended**.

Tutorials start on Week 2

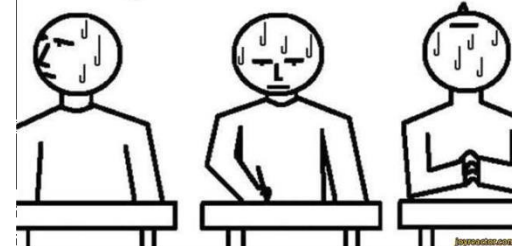
Quizzes start on week 2

Let's make this interactive

The teacher poses a question
and scans the audience...



Must... Not... Make...
Eye... Contact...





	Monday	Tuesday	Wednesday	Thursday	Friday
08:00					
09:00			09:00 12x w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_06 Tutorial 1 09:00 - 10:00 PAR-192-L2-204-Flexible Learning Space (30)		
10:00					
11:00					
12:00					
13:00			13:00 12x w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_01 Tutorial 1 13:00 - 14:00 PAR-115-L7-719-Collaborative Learning Space		
14:00	14:00 12x w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_04 Tutorial 1 14:00 - 15:00 PAR-104-L1-123-Flexible Learning Space (30)		14:00 12x w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_02 Tutorial 1 14:00 - 15:00 PAR-115-L7-719-Collaborative Learning Space		
15:00		15:00 12x w31-39, 41-43 COMP90089_U_1_SM2_2026_L01_01 Lecture 1 15:00 - 16:00 PAR-253-L2-230-Q230 Theatre (196)	15:00 12x w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_03 Tutorial 1 15:00 - 16:00 PAR-115-L7-719-Collaborative Learning Space	15:00 12x w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_05 Tutorial 1 15:00 - 16:00 PAR-142-G-G51-Kimpton Room (30)	
16:00		16:00 12x w31-39, 41-43 COMP90089_U_1_SM2_2026_L02_01 Lecture 2 16:00 - 17:00 PAR-253-L2-230-Q230 Theatre (196)			
17:00					



COMP90089 – Subject Communication

Ed Discussion

- Actively engage by **asking and answering** questions. Peer teaching is the most effective way of learning!
- Please do not post questions too close to an assessment deadline
- (Of course, no assignment solutions should be given away. Doing – or asking for – this is academic misconduct.)

Announcement

- Subject updates, assignment announcement, schedule changes, etc.

Personal/private concerns: Email Daniel

- With private or personal concerns
- Please include COMP90089 in the email subject



COMP90089 - Other Supports

More information on Canvas

- Tutorials location and time
- Academic integrity
- University-wide support student services



Independent Work and Plagiarism

Plagiarism and Misconduct

- The University takes plagiarism **very seriously**, whether deliberate or accidental. **It is your responsibility to understand how to avoid plagiarism and collusion.**
- Details can be found in Canvas.

Academic honesty

- Submitted work are to be done **independently**.
- Please **don't copy your work** from anywhere else.
- If cheating/collusion is detected, both parties – receiver and giver – will be referred to the Faculty of Engineering for handling under the **University Discipline procedures**.



Student Representatives

Student Representative

- Would you like to represent the class on the **student-staff liaison committee**?
- You collect the concerns of your classmates and present them to the committee.
- **Contact me** if you would like to volunteer.



FAQ - About the group project

- Can I pick **my own topic** for the project?
 - Yes, but you need to agree as a group what your topic will be.
- Can I do my project **individually**?
 - No
- What can I do **if my group is not working** for me?
 - Refer to the group contract you will sign by week 3 and please let the instructors know as soon as possible.



Feedback

Feedback

- Throughout the course
- At the end of the semester
 - **Subject Experience Survey (SES)**
 - Constructive criticisms are welcome
 - *Focus on things we can change & are under our control*

Weekly quizzes

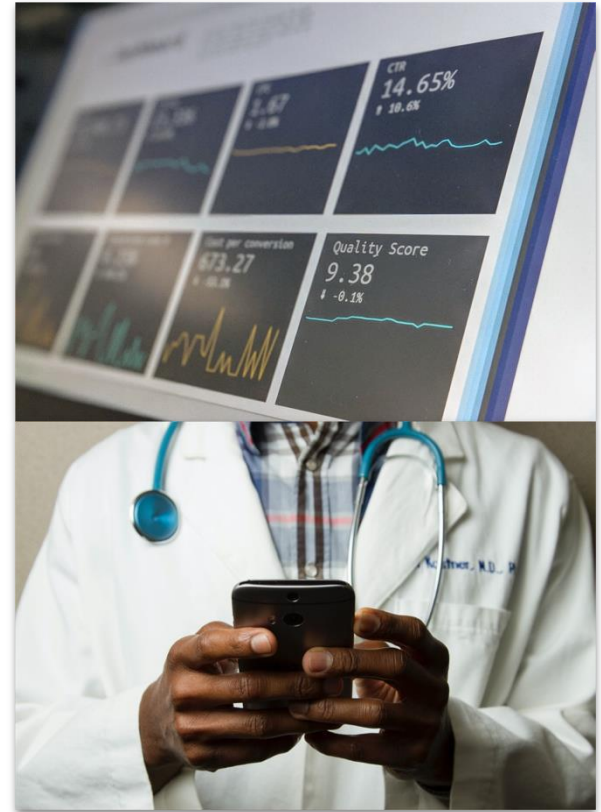
- Review content and your progression





COMP90089 - What to expect?

- Overview of **Machine Learning flavours**
 - Paired with applications in **health**
 - Domain **experts**
 - **No required** ML/healthcare data experience
- Hands on experience with **real clinical data**
 - **MIMIC critical care database**
 - Tutorials
 - Programming assignments
 - Group project



Cogniti

AI learning activities, your way

Cogniti is designed by educators, and built by educators, to empower educators.

Create steerable and accurate agents powered by generative AI. Build mini apps where students can learn interactively. Set up sandboxes where students can build their own.

Cogniti is designed to let teachers build activities that can be given specific instructions, and specific resources, to assist student learning in context-sensitive ways. Because teachers know how learning needs to work.





In-class Activity

Form a group with 2–6 classmates sitting near you and discuss the following:

- What machine learning applications in healthcare have you seen or heard about before?
- What machine learning applications for health would you love to create, something exciting or impactful?



Unique Challenges in Health Data

- While immensely valuable, health data presents several inherent challenges that distinguish it from other data science domains.
- Addressing these requires specialized techniques and careful consideration
 - ✓ **Missing Data:** Up to 30% in typical EHRs, requiring robust imputation strategies.
 - ✓ **Systematic Biases:** Inherited from data collection practices, leading to potential inequities in model performance across patient groups.
 - ✓ **High-Dimensional, Multi-Modal:** Data comes in various formats (numerical, categorical, image, text, time-series), often with a very large number of features.
 - ✓ **Strict Privacy Regulations:** Laws like GDPR and HIPAA necessitate advanced privacy-preserving techniques and ethical data handling.
 - ✓ **Interpretability Needs:** Clinical decisions require ML models that are not just accurate but also explainable and trustworthy.
- These unique challenges require specialized approaches beyond standard ML techniques.



Real-world data set: MIMIC

- Relational database
- Large-scale, freely available
 - >40.000 patients stays in critical care units
 - De-identified health-related database
 - Beth Israel Deaconess Medical Center/US
 - Between 2001 and 2012
- Highly granular data
 - Demographics
 - Vital signs
 - Lab results & procedures
 - Medications
 - Mortality



- **Students are required to obtain access to the MIMIC-IV database**
 - Involves ethics training
 - A tutorial will be provided on Canvas



Python & SQL

- We'll use Python & SQL in our tutorials & assignments
- Anaconda / Miniconda
 - Easy to manage libraries and dependencies
 - Includes many commonly-used scientific libraries
- Python packages
 - <https://www.anaconda.com/download>
 - <https://conda.io/miniconda.html>
- Documentation:
 - <https://conda.io/docs/user-guide/>





Jupyter

Jupyter notebooks (<http://jupyter.org/>)

- Service for **Interactive Computing**
- Mix **code**, **documentation**, and **plots**, interactively
- Built for **data science**
- Great for **reproducible** analysis and exploration
- **We'll use this in our tutorials/assignments**





What do you need to do **this week?**

- **Sign up** for a tutorial slot
- **Get MIMIC-IV access**
 - Google BigQuery
- **Form your group**
- Read **course information** on Canvas
- Complete your **pre-class activities before next week.**

Sneak peek - Lecture 2

